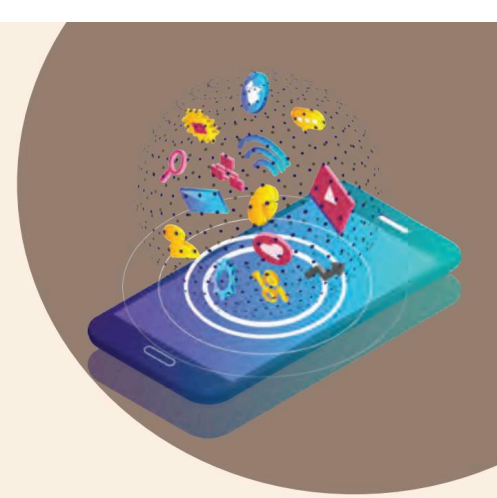


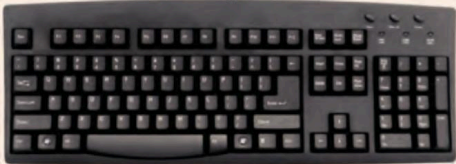
2

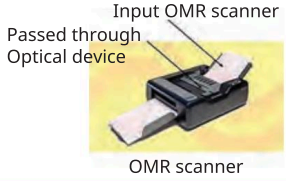
Using Keyboard and Mouse



Data entry is the process of transcribing information or entering physical data into an electronic device, such as a computer, through data entry or input devices.

With a computer, you can use various devices for data entry, as discussed below:

Device Name	Description	Image
Computer Keyboard	Basic input device with keys.	
Mouse	Basic hand-held pointing device.	
Joystick	Used for controlling the movement of a cursor or a pointer on a computer screen.	
Light Pen	A pointing device used to select a displayed menu item or draw pictures on a monitor.	
Magnetic Stripe Reader	Used to read data from the magnetic stripes on most of the banking cards.	

Device Name	Description	Image
Chip Readers	Reads data from the chip on bank cards.	
PIN Pad	An electronic device used to enter data into Automated Teller Machines (ATM) for debit, credit, or smart card-based transactions.	
Optical Mark Reader (OMR)	Used to read and input information from a form.	
Barcode Reader	Scan codes directly from the products, books, and membership cards.	

Among the devices discussed here, the most popular text-based input method is the **keyboard** and the handheld hardware input device that controls a cursor in a Graphical User Interface (GUI) environment is a **mouse**.

To strengthen operational skills while using computers, knowledge of systematic and effective keyboarding skills and typing ergonomics is essential.

Let us understand the basics of the keyboard and how to master keyboarding skills.

Keyboard

A keyboard is one of the peripheral input devices that allow users to communicate with a computer. Communication with the computer is done with the keys or buttons present on the keyboard. A standard keyboard has 104 keys. Each key on a keyboard, when pressed, corresponds to a single character (alphabet, digit, or symbol). Multiple key presses might result in actions or computer commands.



Keyboarding Skills

The ability to touch, type, or input data through the keyboard smoothly while typing is called keyboarding skill. The touch way of typewriting, or touch-typing method, is a form of typing where only fingers are used to strike the keys that must be moved. This method of typing does not require the use of vision. It is a technique that solely relies on the tactile input from the keyboard. The sensation of touch is minimally involved, as this typing technique is controlled by muscle memory developed via extensive practice, using the right typing technique. As a result, the typist's fingers become so accustomed to typing that they automatically press the correct keys without having to look or even feel their way around the keyboard.

Touch typing is done using a standard QWERTY keyboard with the hands placed at a starting location called the '**home row keys**'. The home row keys for the left hand are the ASDF keys and for the right hand are 'JKL'.

Types of Keys

The types of keys on a keyboard can be categorised as:

1. Alphabet Keys
2. Number Keys
3. Punctuation Keys
4. Special Keys
5. Modifier Keys
6. Function Keys
7. Navigation Keys
8. Toggle Keys

Let us discuss each type of keys in detail.

Alphabet Keys

There are 26 alphabet keys on the keyboard. They are marked from A to Z. (Refer Fig. 2.1.)

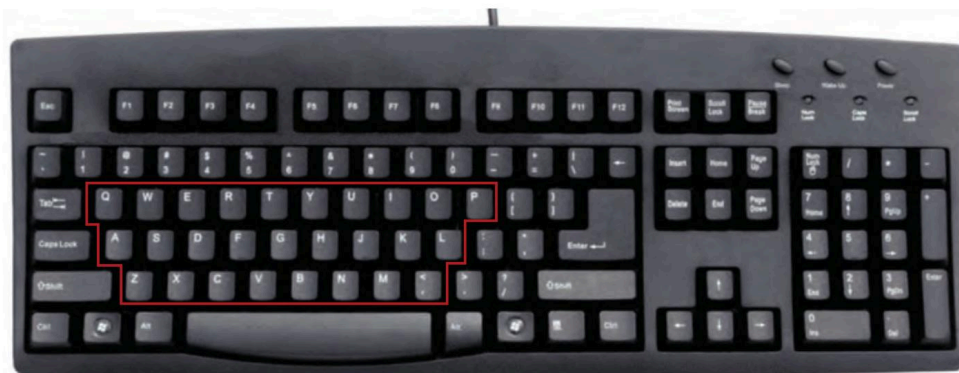


Fig. 2.1: Alphabet Keys on Keyboard

1. You can type words, sentences, and paragraphs with these alphabet keys for documents, articles, and essays.
2. The majority of keyboards feature a QWERTY layout with keys arranged in a non-alphabetical manner.
3. Having the conventional English-language key layout that starts with the letters Q-W-E-R-T-Y, reading from left to right across the top row of the keyboard, is known as a QWERTY keyboard. It can be found on computers, typewriters, laptops, or other electronic devices.

Number Keys

There are 10 number keys on a keyboard, numbered from 0 to 9. There are two separate positions and functions for the number keys, as shown in Fig. 2.2.



Fig. 2.2: Number Keys on a Keyboard

Position I: Symbol Keys

The keys with symbols on them are located above the letters on the keyboard. When pressed individually, these keys function as number keys. However, when used in combination with the shift key, they function as symbol keys.

The following table shows the symbol given above the digits (Position I):

Number Keys	Respective Symbols
1	!
2	@
3	#
4	\$
5	%
6	^
7	&
8	*
9	(
0	)

Position II: Numeric Keypad

It is located at the end of the right side of the keyboard. It is also known as a numeric keypad since it resembles a telephone number pad and used for entering numbers at a faster pace. It works in two different modes. The Num Lock key is provided to switch between these two modes.

- Num Lock On:** On pressing the numeric keys, the respective digits are typed.
- Num Lock Off:** When Num Lock is off, the keys 8, 2, 4, and 6 function as arrow keys for navigation, whereas 7, 1, 9, and 3 act as home, end, page up, and page down keys, respectively.



Did You Know?

Usually, some of the keyboards on laptops do not have a numeric keypad on the right side.

Punctuation Keys

Correct punctuation adds clarity and precision to the text. There are generally 14 punctuation marks used in the English language. (Refer Fig. 2.3.)

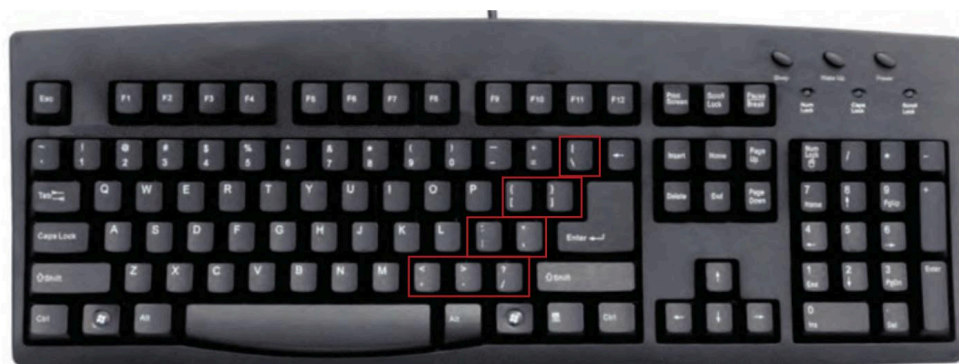


Fig. 2.3: Punctuation Keys on Keyboard

The following chart contains the punctuation mark's name, symbol, and examples:

Punctuation Mark Name	Type of Punctuation Mark	Example
Period	.	The cat is under the table.
Question Mark	?	Did you complete your homework?
Exclamation Mark	!	What a lovely view you have here!
Comma	,	In both positions, there are 10 keys, ranging from 0 to 9.
Colon	:	You have two choices: finish the work today or lose the contract.
Semicolon	;	My daughter is a teacher; my son is a doctor.
Dash	En dash (-) & Em dash(—)	Princeton–New York trains (En dash) She gave her answer—No! (Em dash)
Hyphen	-	Son-in-Law
Brackets	[]	I read that novel [The Great Gatsby] in high school.
Braces	{ }	$21\{11+[3-2]\}=x$
Parentheses	()	I am going to visit my grandma (my dad's mom) today.
Apostrophe	'	She's always on time.
Quotation Mark	" "	"Raja thinks that his brother [Akshit] will be a great dad", said Vishal.
Ellipsis	...	'Today ... we vetoed the bill'.

Special Keys

The special keys present on the keyboard are used to carry out special actions in the text. (Refer Fig. 2.4.)

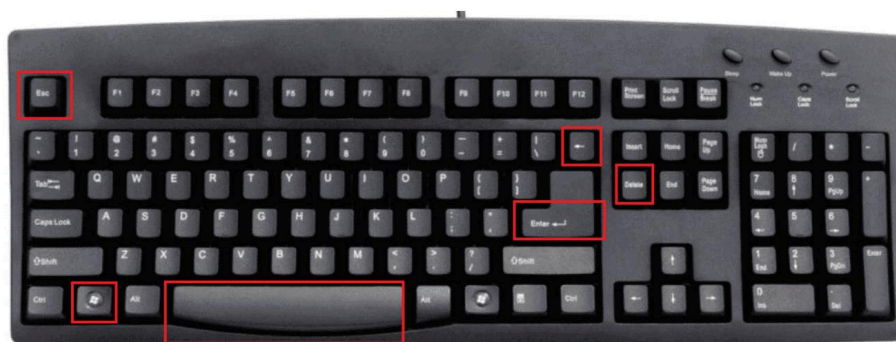


Fig. 2.4: Special Keys on a Keyboard

Some of the special keys and their functions are listed below:

Enter or Return key It is used for entering commands or advancing the pointer to the start of the subsequent line.

Esc key The Escape key is used to end (or “escape”) applications and tasks as well as communicate unique codes to devices.

Spacebar key It is used to add space between two independent words to form a sentence.

Backspace key It is used to delete the words and spacing before the cursor.

Windows key It is used to open the Start Menu and use different shortcuts, like Windows + I, to open the Settings app.

Delete key The Delete key does not move the pointer; instead, it deletes the character that is at the current cursor position of the currently selected object.

Modifier Keys

A modifier key is a key that modifies the action of another key when pressed simultaneously. These are used in combination with other keys to perform any action.

Shift Key It is used as a keyboard shortcut key in conjunction with another key to switch between capital and small letters. (Refer Fig. 2.5.)

Ctrl Key To create control characters, combine the control key with other keys. Depending on the program that is running, each control character has a different meaning.

Alt Key This key, known as an Alternate, functions as a second control key. In addition, it is combined with other keys as a keyboard shortcut key.

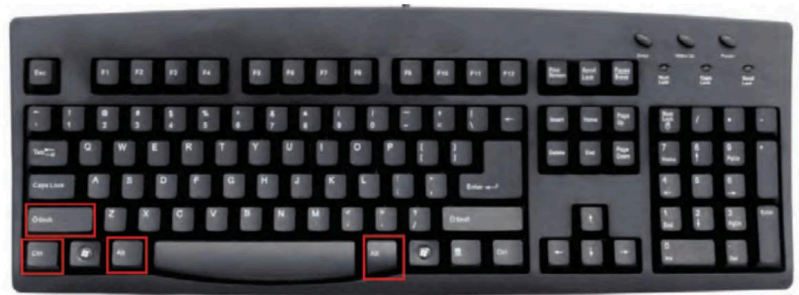


Fig. 2.5: Modifier Keys on Keyboard

Function Keys

The top row of the keyboard is home to the function keys, which start with the letter F, from F1 to F12 (refer Fig. 2.6), on the keyboard are the function keys. Depending on the software that is being run, these keys have varied meanings.

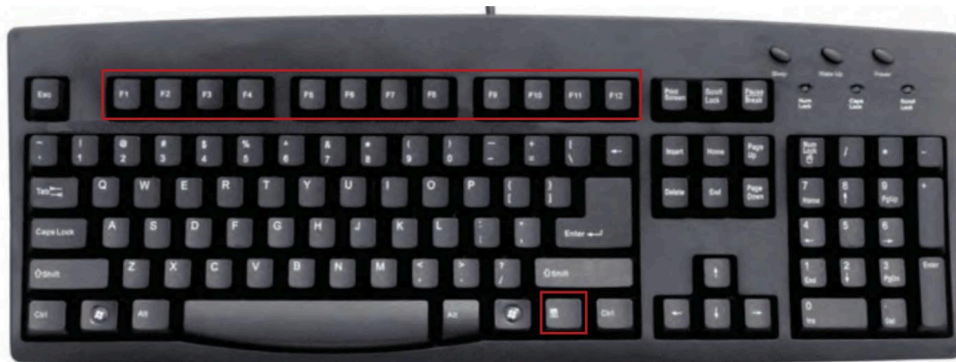


Fig. 2.6: Function Keys on Keyboard

Navigation Keys

The navigation keys are the keys on a keyboard used to move around a screen. (Refer Fig. 2.7.) These may include up, left, right, and down arrows. These allow you to navigate through documents or other information.



Fig. 2.7: Navigation Keys on Keyboard

1. The numeric keypad keys 8, 4, 2, and 6 can also be used as navigation keys when Num Lock is off.
2. These keys are very useful when playing games or handling tasks like video editing.

Toggle Keys

By pressing a toggle key, a user can turn something on or off. (Refer Fig. 2.8.) The Caps Lock, Num Lock, and Scroll Lock keys are commonly referred to as toggle keys.

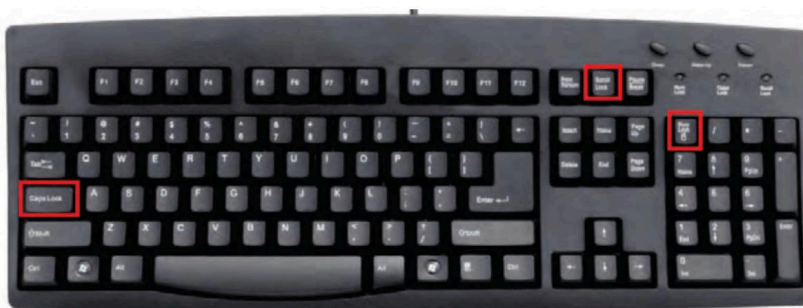


Fig. 2.8: Toggle Keys on a Keyboard

Caps Lock key All alphabetic characters are capitalised when the setting is **ON**; otherwise, they are all small.

Scroll Lock key This key is no longer seen on the majority of contemporary keyboards because it is rarely used. However, if it is available on a keyboard, it aids in stopping page scrolling until it is in use.

Num Lock key This key is used to turn numeric 1–9 keys off. With this key in on mode, the numeric keys 8, 4, 2, and 6 can be used as navigation keys.

Remember

There is an electrical contact made when a key is pressed. These electrical signals are sent in coded form to a microcontroller, which then relays them to the computer, describing the character that matches that key.

Home Keys and Guide Keys

With good keyboard skills, time is saved, tiredness is reduced, text creation speed is increased, accuracy is improved, and workplace productivity is increased. These good keyboarding skills can be achieved only by learning the touch-typing method. Touch typing is a way of typing (with all your fingers) without looking at the keyboard; only the movement of fingers is required to press keys. With extensive training, touch typists use muscle memory to remember where the keys are present on the keyboard.

The keys are placed as:

1. Home row keys and
2. Guide keys

A, S, D, and F are the **left-hand home row keys** on a typical QWERTY keyboard, whereas **J, K, L, and ;** are the **right-hand home row keys**. The fingers are trained to make the correct movement to other keys, and each finger returns to its home key instantly.

The keys '**F**' and '**J**' are known as the **left- and right-hand guide keys**, respectively. Both have a small and raised noticeable mark that allows the touch typist to correctly place their fingers on the home keys.

The following points should be kept in mind for successful keyboarding techniques:

1. Press the keys with a feather touch, and do not put undue pressure on them.
2. Rest your fingers on the home row while typing.
3. Allow your fingers to fall naturally on the keys so that each rest on top of the next key along the same horizontal row.

4. While typing, release the key immediately as soon as you press it. Holding the key for a long time will repeatedly type the same character.
5. Proper sitting arrangement improves the typing speed and reduces stress on the hands.

Guide for Typing

A cursor is a small blinking line that flickers in a sentence while typing and shows the position at which you are. It serves as a guide for typing.

Positioning of Fingers on the Keyboard

Nowadays, proficiency on a computer keyboard is similar to mastery of a pen or pencil.

Mastery of typewriting skills involves gaining the requisite knowledge and skills of keyboard operation through proper finger positioning.

A touch typist starts by placing his or her fingers on the 'start position' on the middle row and knows which finger to move and how much to move to reach any required key. It is important to learn to place fingers in the starting position blindly, as the hands are frequently raised from the keyboard to operate the computer mouse.

The fingers of both hands must operate the keys designated to them on each row to learn how to use the keyboard. The keyboard learning process begins from the second row (Home Row), followed by the third row (Upper Row), the first row (Bottom Row), and then the fourth row (Number Row).

Allocation of Keys to Fingers on the Second Row (Home Row)

The home row keys are A, S, D, F, G, J, K, L, and ;. It is recommended to always keep your fingers on or close to the keys. Each hand and finger will only serve a range of predefined keys. The thumbs will rest on the space bar. Regular practice will build the muscle memory necessary to increase speed and accuracy.

The **left hand** should focus on the following home row keys:

1. The pinkie finger should be on key **A**.
2. The ring finger on **S**.
3. The middle finger on **D**.
4. The index finger should be on **F**.

The **right hand** should focus on the following home row keys:

1. The index finger should be on **J**.
2. The middle finger should be on **K**.
3. The ring finger should be on **L**.
4. The pinkie should be on **;**.

The touch typist places the eight fingers in a horizontal row along the middle of the keyboard (the home row) and places the two thumbs on the space bar. The remaining two keys 'g' and 'h' on the second row are operated by the forefingers (index finger) of the left and right hands, respectively.

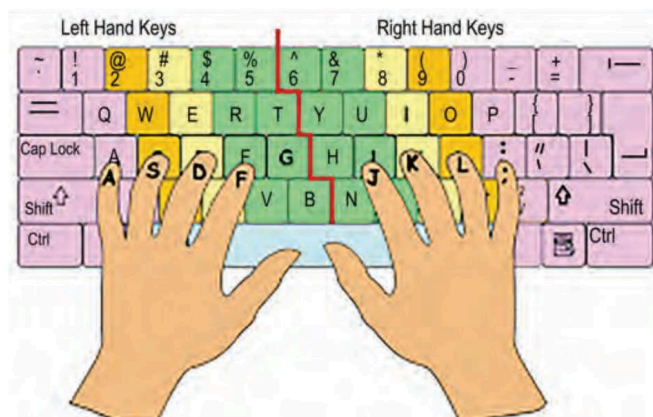


Fig. 2.9: Left- and Right-Hand Keys

Allocation of Keys to Fingers on the Third Row (Upper Row)

The upper row keys are Q, W, E, R, T, Y, U, I, O, and P. It is recommended to always keep your fingers on or close to the keys. Each hand and finger will only serve a range of predefined keys. Regular practice will build the muscle memory necessary to increase speed and accuracy.

The **left hand** should focus on the following upper-row keys:

1. The pinkie finger should be on key **Q**.
2. The ring finger on **W**.
3. The middle finger on **E**.
4. The index finger should be on **R**.
5. The index finger should be on **T**.

The **right hand** should focus on the following upper-row keys:

1. The index finger should be on **Y**.
2. The index finger should be on **U**.
3. The middle finger should be on **I**.
4. The ring finger should be on **O**.
5. The pinkie should be on **P**.

Keep your fingertips on the home row while typing on the upper row. Lifting the left hand's fingers, type q w e r t (r and t are typed by the same finger) by pressing the upper row key one at a time.

Now, raise your right hand's fingers once again, press the upper row keys one at a time, and type p o i u y (u and y are typed with the same finger).

One finger should be moved at a time, while the other fingers should stay in the home row.

Allocation of Keys to Fingers on the First Row (Bottom Row)

The Bottom row keys are Z X C V B N M. All eight fingers on both hands were utilised to press the keys in the first two rows.

The **left hand** should focus on the following bottom-row keys:

1. The ring finger on **Z**.
2. The middle finger on **X**.
3. The index finger should be on **C**.
4. The index finger should be on **V**.

The **right hand** should focus on the following bottom-row keys:

1. The index finger should be on **B**.
2. The index finger should be on **N**.
3. The middle finger should be on **M**.
4. The ring finger should be on ,
5. The pinkie should be on .

Remember

The little finger of the left hand is not used when typing on the first row, meaning that it does not have a key assigned to it.

Allocation of Keys to Fingers on the Fourth Row (Number Row)

The **left hand** should focus on the following fourth-row keys:

1. The pinkie should be on **1**.
2. The ring finger on **2**.
3. The middle finger on **3**.
4. The index finger should be on **4**.
5. The index finger should be on **5**.

The **right hand** should focus on the following fourth-row keys:

1. The index finger should be on **6**.
2. The index finger should be on **7**.
3. The middle finger should be on **8**.
4. The ring finger should be on **9**.
5. The pinkie should be on **0**.

Using a Numeric Keypad

To enter numerical data correctly, the numeric keypad must be used efficiently. There are four columns and five rows on the numeric keypad. Home Row is the grouping of 4, 5, 6, and +. This is the row that a touch typist starts by practicing.

The guide key on a numeric keypad is 5, and it has a little raised, evident mark that the touch typist might use as a reference when placing their fingers on other keys. The right-hand thumb is used to press the remaining keys on the numeric keypad, starting with '0'. The calculator keyboard can also be used to practice using the numeric keypad.



Fig. 2.10: Positioning of Fingers on Numeric Keypad

Pointing Device

A pointing device is a sort of input device that allows a user to interact with a computer by moving a pointer across a screen to choose icons and perform desired actions. The mouse, which was designed by Douglas England in 1967, is the most basic pointing device for desktop computers.

In addition to typing directly, we can select menu items from the computer screen, such as the Save or New options. This can be accomplished with point-and-draw devices such as a mouse. Such devices can be used to draw basic shapes, block arrows, and charts. Joysticks, light pens, and touch screens are further examples of point-and-draw devices.

Mouse

A mouse is a small handheld input device that allows you to move around the screen and enter commands without touching the keys. Mouse can be wired or wireless. The mouse is an essential component

Remember

The other pointing device is the touchpad for the laptop, and a small number of users prefer trackballs over the mouse.



Did You Know?

A wireless mouse uses infrared or radio waves to communicate with the computer.

of any PC's graphical user interface (GUI). The mouse gets its name from its resemblance to a short, corded, elliptical gadget that resembles a mouse tail.

The mouse is set up by default to work with the right hand. Left-handed users can adjust the settings as per the requirement. Put your right hand on the mouse with your index and middle fingers on the left and right buttons, respectively. Use your ring finger and thumb to grip the mouse. Lightly push one mouse button, then quickly let go of it to click.

Types of Mouse

There are varieties of mouse used with computers. For a desktop computer, the most common type of mouse is an optical mouse that connects to the USB (Universal Serial Bus) port and is known as a USB mouse. For laptop computers, the most common type of mouse is the touchpad. Following is a list of the different types of mouse that can be used in computers:

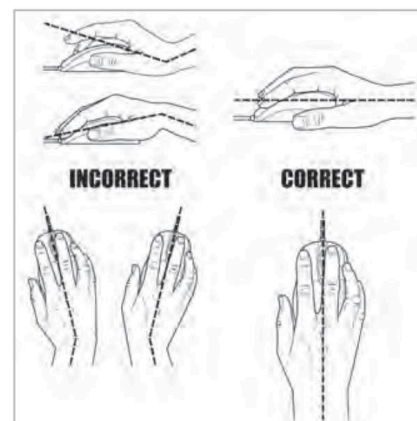
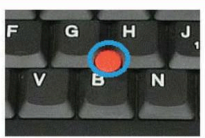


Fig. 2.11: Holding a Mouse

Optical	
Mechanical	
Cordless (Wireless)	
Foot Mouse	
Touchpad (Glide point)	
Trackball	
TrackPoint	

J-Mouse	
IntelliMouse (Wheel Mouse)	
Laser Mouse	

Advantages of Using a Mouse

Following are the advantages of using a mouse.

Accuracy For tasks like clicking small objects and selecting text, a mouse is more precise than other input devices like keyboards.

Speed A mouse can be used to navigate the graphical user interface faster than other input devices.

Comfort A mouse is more comfortable to use over a longer period than other input devices.

Mouse Pointer

A mouse pointer allows pointing on the screen. This mouse pointer appears as you move it around the screen. When using a web browser, the pointer changes to a pointing finger symbol as it moves across the page. This is used to identify a hyperlink. To navigate to another page, use the left mouse button once.

Basic Mouse Actions

Following are the basic mouse actions:

Click or Left-click This is used to select an item. Press the left button once with your index finger.

Double-click It is used to start a program, open a file, or trigger an action.

Right-click The right button on a mouse is typically used to provide additional information and/or properties of an item selected. For example, if you highlight a word in Microsoft Word, pressing the right button will display a drop-down menu containing the options to cut, copy, paste, change the font, etc.

Drag and Drop A drag-and-drop activity requires students to drag a piece of text or an image to one or more corresponding drop zones. Drag-and-drop activities are best utilised when students are engaged in sorting, grouping, matching, or placing information.

Scroll A little wheel placed between the left and right mouse buttons on your mouse may be visible. This is referred to as the scroll wheel. You can go up and down a document or webpage by rotating the scroll wheel with your index finger.

Selecting/Blocking It is another technique to pick text that has to be changed or formatted. You must click at the beginning of the word or sentence and hold down the left mouse button while dragging along the text, which will be highlighted in black. At the end of the paragraph or sentence, release the button.

Typing Ergonomics

The science of typing ergonomics focuses on utilising good posture and stretching practices when using a computer to prevent frequent occupational injuries. It offers logistical support for speedy and accurate typing.

Sitting Posture

1. The height of the chair should be adjusted in such a manner that your feet are flat on the floor.
2. The hips should be slightly higher than the knees.
3. Keep your back straight.
4. The back of the chair should be somewhat inclined at a 100- to 110-degree angle.
5. Ensure that the keyboard is close and directly in front of you.



Fig. 2.12: Correct and Incorrect Sitting Posture

Hands Position

1. Put your forearms at a level with the keyboard and your palms down.
2. Elbows should be slightly higher than wrists when operating keys or a mouse. Neither touch the elbows to the body nor be too far away from the body.
3. Bend your elbows at a 90-degree angle.
4. The desk should be approximately 680- to 720-mm high.



Fig. 2.13: Hands Position

Think and Tell

- Do you adopt the proper seated position when using a computer?
- Are your workstation and chair ergonomic?
- If not, have you made any necessary adjustments?
- Are the positions of your keyboard, mouse, and computer table correct?

Yes/No
Yes/No
Yes/No
Yes/No

Activity Time

Guidelines to attempt Activity 1 and 2

- Practise typing the character without looking at the keyboard.
- Arrange the letters so that they are stacked one on top of the other.
- Focus on the text.
- Type slowly and accurately.
- Repeat this exercise until you have mastered it.

Activity 1: Get Perfection Over Left Hand

(Individual Work)

Type the following in double-line spacing to get perfection over the **Left**-hand Home Row. (Using letters A, S, D, and F)

```
ffffdddddssssaaaffffdddddssssaaaffffdddddssssaaaa  
ff dd ss aa ff dd ss aa ff d ff d ff s ff a ff a  
fdsafdsafdsafdsafdsafdsafdsafdsa  
asdfasdfasdfasdfasdfasdfasdfasdf  
fads dafssa fdds affasd ad fssd afafdsfdsa
```

Activity 2: Get Perfection Over Right Hand

(Individual Work)

Type the following in double line spacing to get perfection over the **Right**-hand Home Row. (Using letters J, K, L, and punctuation ;)

jkkklll ; jkkklll ; jkkklll ;jj kk ll ; jj kk ll ;
jj kk ll ; jj kk ll ; jj kjj ljj ; jkl ; jkl ; jkkkll ;
jkl; jkl; jkl; jkl; jkl; jkl; jkl; jkl; jkl; jkl;
|kl; |kl; |kl; |kl; |kl; |kl; |kl; |kl; |kl; |kl; |kl;

Chapter Checkup

A Select the correct option.

1. _____ key is not an example of modifier keys.
a Alt **b** Ctrl **c** Spacebar **d** Shift
2. A touch typist starts by placing his or her fingers on the 'start position' in the _____.
a Left Row **b** Middle Row **c** Right Row **d** Down Row
3. To type capital letters, switch ON the _____ by pressing the key before typing.
a Esc Key **b** Caps Lock Key **c** Enter Key **d** Alt or Ctrl Key

B Fill in the blanks with the most suitable words.

- 1 On the numeric keypad, 0 is to be pressed by the _____ thumb.
- 2 The keys F and J are known as _____.
- 3 The _____ key is used to delete the character on the left-side of the cursor.
- 4 Q, W, E, R, T, Y, U, I, O, P are known as _____ row keys.

C State whether the following is True or False. Correct the statements that are false.

- 1 The function keys have different meanings in different software.
- 2 On a numeric keypad, the number 8 is the guide key.
- 3 The control key is used in conjunction with other keys.
- 4 The Alt key is always used with another key.

D Answer the following questions. (Solved)

Q1. Who is a touch typist?

A1. A touch typist is someone who types without looking at the keyboard. Muscle memory helps the user remember where the keys are available. Touch typists position their fingers in the centre of the row and know which finger to move and how much to move to reach any key.

Q2. Differentiate between Home Keys and Guide Keys.

A2. Home Key – The home keys for the left hand are A, S, D, and F, and the home keys for the right hand are ; (semi-colon), L, K, and J.

Guide Key – The keys 'F' and 'J' on a computer keyboard are known as left- and right-hand guide keys, respectively.

Q3. Sneha is not aware of typing ergonomics. Explain it to her by describing about the sitting posture.

A3. The science of typing ergonomics focuses on utilising good posture and stretching practices when using a computer to prevent frequent occupational injuries. It offers logistical support for speedy and accurate typing.

For the sitting posture:

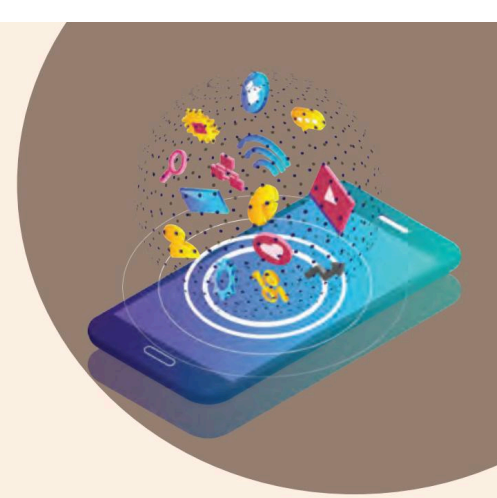
- The height of the chair should be adjusted in such a manner that your feet are flat on the floor.
- The hips should be slightly higher than the knees.
- Keep your back straight.
- The back of the chair should be somewhat inclined at a 100- to 110-degree angle.
- Ensure that the keyboard is close and directly in front of you.

Answer Key

- A** 1. c 2. b 3. b
- B** 1. Right 2. Guide keys 3. Backspace 4. Upper
- C** 1. True.
2. False. The guide key on a numeric keypad is 5
3. True.
4. True.

3

Using Typing Software



Keyboarding is a technical skill that everyone should know. What makes a real difference is how good you're at typing, i.e., speed and accuracy. If you do touch type at an above-average word-per-minute rate, you can save hundreds of hours per year. By devoting some hours to learn touch typing, every student can equip themselves with an essential skill that will pay off time and time again for the rest of their life. Effective typing is necessary for any office work.

Typing software is a computer program that teaches users how to type faster and more accurately, typically through typing drills and lessons. Although there are some programs for skilled users, most typing software is designed for beginners.

Many of the software programs available today are free of cost and are customised to learn typing efficiently and effectively. Examples of typing software are Rapid Typing Tutor, Typing Instructor, The Typing Cat, Typing Master, etc.

Touch Typing

Touch typing is a method of typing without using your vision. The typist uses muscle memory to find keys on the keyboard without having to look at them. It involves placing all eight fingers on the home row and using them to locate the keys by touch, allowing for more efficient typing. Through extensive training and practice, individuals can develop muscle memory and improve their typing speed and accuracy. Touch typing is indeed a valuable skill that can be honed through various exercises and workouts, leading to enhanced professional keyboard skills.

Benefits of Touch Typing

Speed Increase in speed and efficiency is one of the most alluring benefits of learning touch typing.

Accuracy Learning to touch type will help you type more correctly and accurately. No one would comprehend what you were trying to say if you were typing quickly but incorrectly.

Time Doing the assignment within the allotted time is another advantage of touch typing. Less time will be needed to complete the work if the typing speed is fast.

Focus You don't have to search for the keys while typing if you know touch typing. Learning touch typing allows you to focus on your work, thereby increasing productivity.

Requirements of Touch Typing

Following are the requirements for touch typing:

Hardware A computer system with a keyboard and mouse.

Operating System Software Windows 7 and above

Application Software Rapid Typing Software 5.4

Rapid Typing Tutor Software

Rapid Typing Tutor is a free and open-source Software (FOSS) program designed to help users improve their typing skills. It offers lessons, games, and typing tests to assist users in learning and practising touch typing. Rapid Typing Tutor is a convenient and easy-to-use keyboard trainer that will help you improve your typing speed and reduce typing mistakes. With its lessons organised for various student levels, it will teach you touch typing or enhance your existing skills.

Available Courses While Learning Rapid Typing

Rapid typing offers a variety of courses based on difficulty level, including Introduction, Beginner Experienced, Advanced, and Testing.

Introduction Course This course covers every character on your keyboard.

Beginner Course The key placement on a keyboard can be learned with the aid of this course. After completing this course, you won't need to look at the keyboard to know which finger to press each key. Capital letters, special symbols, and the numeric pad are all covered throughout the course.

Experienced Course This will assist the user in increasing typing speed by learning commonly used syllables and words.

Advanced Course By entering actual text, the user can practice their typing skills with the help of this course.

Testing Course With the help of an advanced typing test in this course, the user can assess his present typing speed and accuracy and print a copy of the results.

Starting Rapid Typing Software

Once you have downloaded the software, follow the given steps to start a rapid typing software:

1. Click the **Start** button > **All apps** > Select **RapidTyping 5** from the list. OR

Type **RapidTyping 5** in the search box and hit the **Enter** key. The screen is displayed as shown in Fig. 3.1.

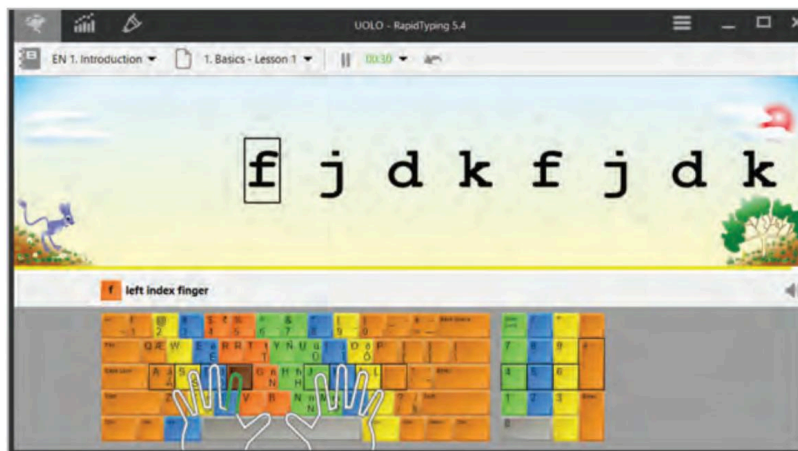


Fig. 3.1: Rapid Typing 5.4

Components of Rapid Typing Tutor Software

Following are the components of the Rapid Typing window:

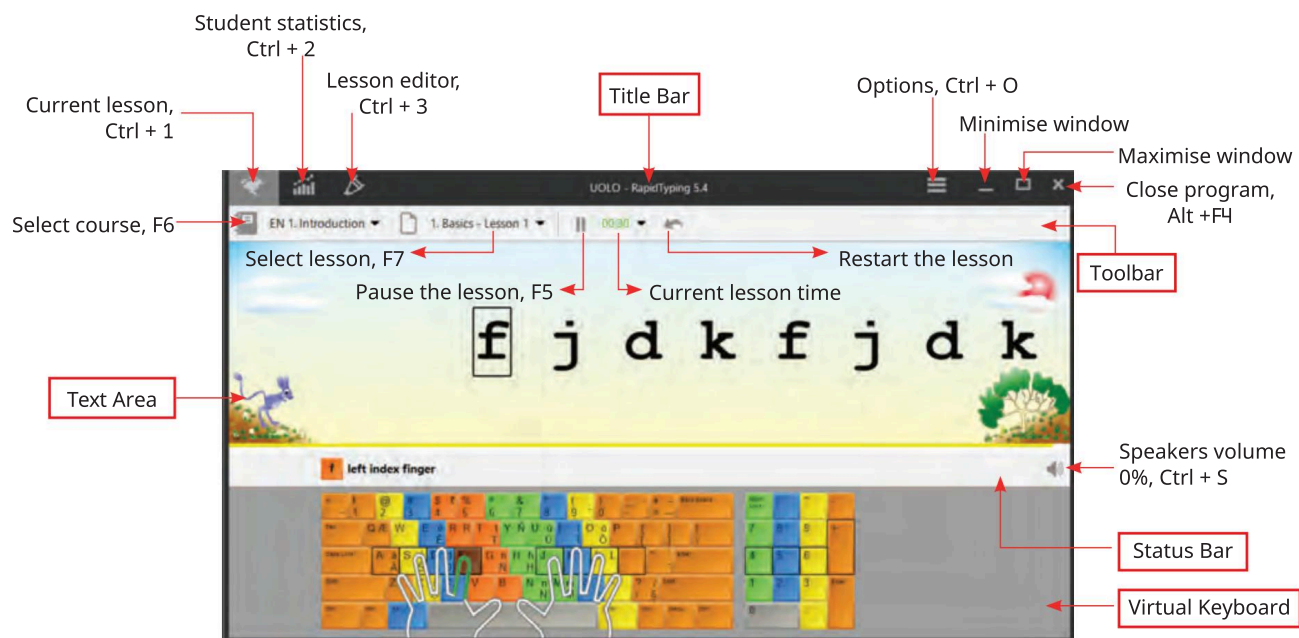


Fig. 3.2: Components of RapidTyping Software

Title Bar: The title bar is the top-most bar of the RapidTyping window.

The top-left corner contains three buttons for moving between basic program modes:

1. **Current Lesson:** Used to check the current lesson.
2. **Student Statistics:** Used to check the statistics of the student's typing skills.
3. **Lesson Editor:** Used to edit the lesson settings.

The top-right corner contains the following buttons:

1. **Options:** Used to change the settings such as language, background, text colour, themes, sound, font, help, update, and log out of the RapidTyping software.
2. **Maximise, Minimise, and Close Buttons:** The maximise and minimise buttons are used to control the window size and the close button is used to close the window.

Toolbar: The toolbar allows you to select the course and lesson, as well as customise the primary program features. The **Lesson Control section** allows you to start/stop the current lesson, return to the beginning, and view the current speed, accuracy, and remaining or elapsed lesson time in real time.

Text Area: The text to be typed appears in the text area in the centre of the window.

Status Bar: The status bar at the bottom contains suggestions for the current lesson. It provides tips for the user to follow while typing. The progress bar contains the lesson length and metronome. The volume control is used to increase or decrease the sound generated by keys while typing.

Virtual Keyboard: This will assist you in learning to type with all ten fingers. In the keyboard settings, you can customise its options.

How to Start a Lesson

Once you open RapidTyping software, let's start practising on the typing software by following the given steps:

1. Click the **Current Lesson** button tab. Generally, it is selected by default. You will see two drop-down lists see Fig. 3.3.

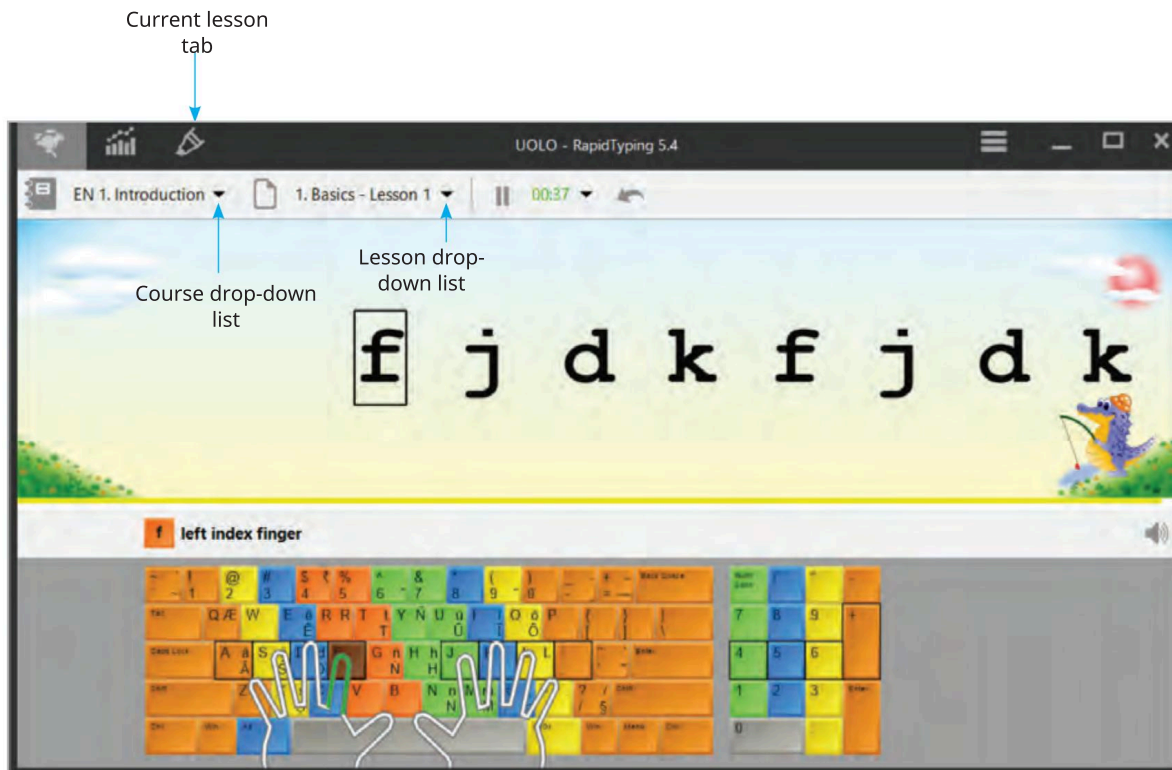


Fig. 3.3: Selecting the Current Lesson

Depending on your knowledge and experience, select a course as shown in Fig. 3.4.

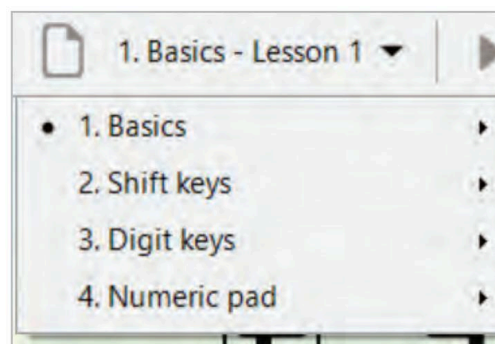


Fig. 3.4: Types of Courses

2. After selecting the course, select the lesson you want to learn. Lessons are categorised by function, for example, **Basics**, **Shift keys**, **Digit keys**, and **Numeric pad**. (See Fig. 3.5.) These parts are included in the introduction and beginner courses.

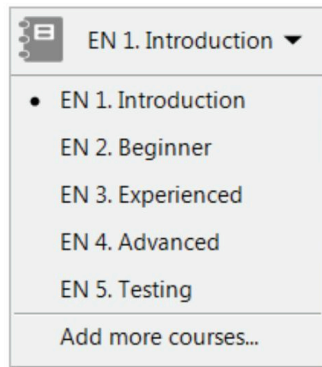


Fig. 3.5: Types of Lesson

3. To begin with, first select the course, **Introduction**, and then select the lesson, Lesson 1. (See Fig. 3.6.)

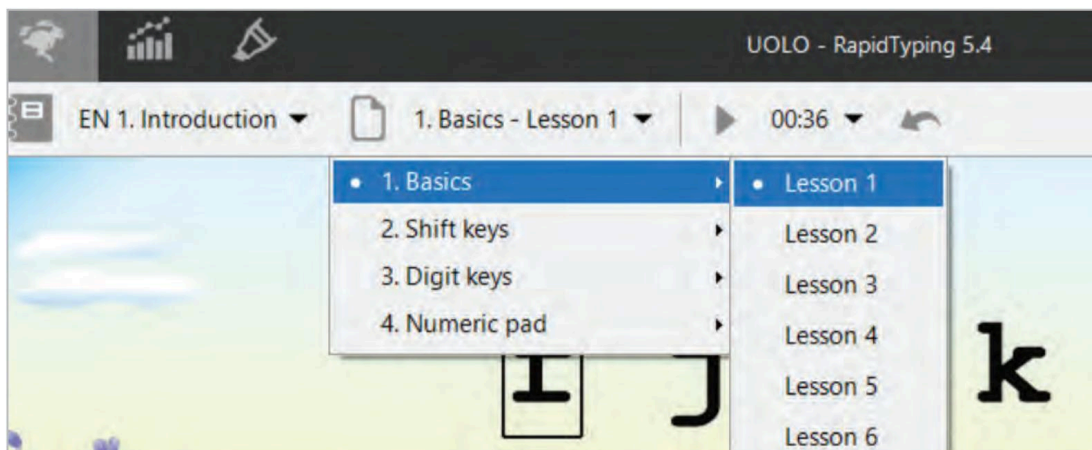


Fig. 3.6: Starting Lesson 1

4. Put your fingers in the base position, as shown on the keyboard (Fig. 3.7). When everything is ready, press any key to begin the lesson. Now, use the keyboard to enter the text or letter that is displayed in the text area.

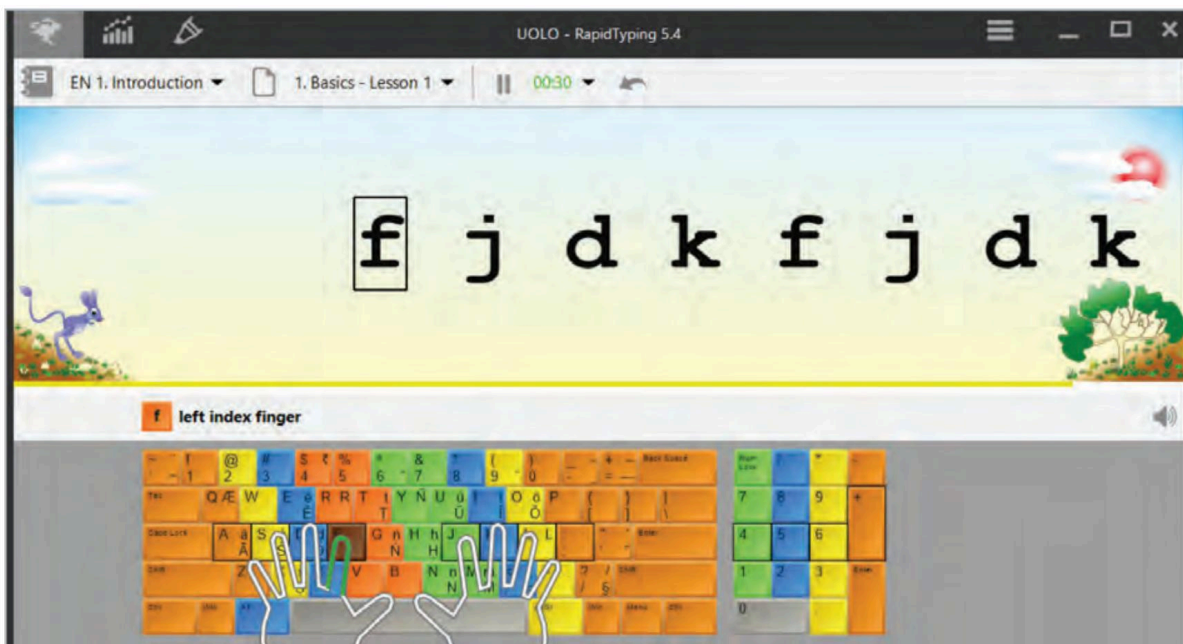


Fig. 3.7: Placing Fingers on the Virtual Keyboard

5. While you are typing, see the current speed, accuracy, and remaining lesson time in real time. (See Fig. 3.8.)

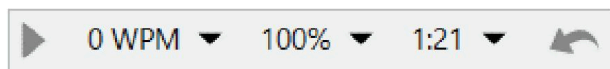


Fig. 3.8: Current Speed, Accuracy, and Remaining Lesson Time

6. To begin a lesson, either click the  button or press any key. To pause a lesson, click the  button that is displayed or press the F5 key. To restart the current lesson, click the  button, or press the F8 key.
7. You can also specify time remaining or time elapsed here, as well as the speed and accuracy indicators for real-time controls.
8. The bottom text area contains the lesson status bar. It recommends which fingers to use to type the current character. Also, it shows two progress bars, which indicate the proportion of the current lesson that has been completed in the green progress bar (below), and the upper yellow progress indicator shows the acceptable time for typing one character. (See Fig. 3.9.)



Fig. 3.9: Status Bar

9. Each lesson ends with the appearance of the Results window, as shown in Fig. 3.10. The Results pane shows your successes and provides several additional possibilities.



Fig. 3.10: Result Overview

Typing Speed

Various accuracy levels can be used to determine the typing speed.

1. How many characters are typed for a given period?

Remember

A virtual keyboard will assist a beginner in locating the required keyboard characters. While typing, you should establish and keep a rhythm. This indicates that there should be consistent spacing between keystrokes.

Remember

Rapid Typing reports this as a slowdown error when the yellow progress bar runs out. In the Course Options, Metronome section, you can customise it.

2. How many keystrokes are done for a given period? and
3. How many words are typed for a given period (the least accurate)?

Gross speed and **net speed** are the metrics used to calculate an operator's typing speed.

Gross or raw speed is a measurement of how quickly you type without accounting for errors, and net speed is a calculation of how quickly you can type error-free.

The following lists the detailed descriptions of types of typing speed.

1. **Words per minute (wpm):** The number of words typed in one minute.
 - Gross speed = WPM = (Words without errors + Words with errors) / Time spent in minutes
 - Net Speed = Net WPM = WPM – (Words with errors / Time spent in minutes)
2. **Character per minute (cpm):** The number of characters typed in one minute.
 - Gross speed = CPM = (Characters without errors + Characters with errors) / Time spent in minutes
 - Net Speed = Net CPM = CPM – (Characters with errors / Time spent in minutes)
3. **Keystrokes per minute (kpm):** The number of keystrokes in one minute.
 - Gross speed = KPM = (Keystrokes without errors + Keystrokes with errors) / Time spent in minutes
 - Net speed = Net KPM = KPM – (Keystrokes with errors / Time spent in minutes)

For example

1. A typist typed 480 characters per 4 min with errors in 40 characters.
 - Gross speed = 480 characters / 4 min = 120 cpm
 - Net speed = 120 cpm – (40 errors / 4 min) = 110 net cpm
2. A student typed 450 words per 10 min with errors in 50 words.
 - Gross speed = 450 words / 10 min = 45 wpm
 - Net speed = 45 wpm – (50 errors / 10 min) = 40 net wpm

Typing Accuracy

Typing accuracy is measured as the proportion of correctly typed characters to total characters.

The various formulas for calculating typing accuracy are listed below:

Percentage of Word Accuracy

Accuracy% = (100% – (Words with errors * 100%) / Total number of words)

Percentage of Character Accuracy

Accuracy% = (100% – (Characters with errors * 100%) / Total number of characters)

Percentage of Keystroke Accuracy

Accuracy% = (100% – (Incorrect keystrokes * 100%) / Total number of words)

Typing Errors

Typing inaccuracy is the percentage of wrongly typed words out of the total number of typed words. The various formulas for calculating **Errors** are listed below:

Percentage of Word Errors

Errors % = (Words with errors * 100%) / Total number of words

Percentage of Character Errors

Errors % = (Characters with errors * 100%) / Total number of characters

Percentage of Keystroke Errors

Errors % = (Incorrect keystrokes * 100%) / Total number of keystrokes

Typing Rhythm

The typing rhythm is highly crucial in touch-typing approaches. Typing rhythm implies that keystrokes should be spaced evenly. The slowdown indicator is used to manage consistent typing speed.

Percentage of Slowdown in the Words

Slowdown % = (Words with delay * 100%) / Total number of words

Percentage of Slowdown in the Characters

Slowdown % = (Characters with delay * 100%) / Total number of characters

Percentage of Slowdown in the Keystrokes

Slowdown % = (Keystrokes with delay * 100%) / Total number of keystrokes

Understanding Result Window

Understanding results is a crucial task; it helps you understand where to improve. Once you complete a lesson, the result window will display the overall rating. It will track your progress by displaying typing speed, accuracy, and keystroke slowdown. Also, there are three tabs present in the lower portion of the window, namely, Next step, Detailed statistics, and Errors overview.

Next Step

The **Next Step** tab advises that you take: or move on to the next lesson; compare your outcomes once more to the course objectives in the course selections. If you meet the three requirements (speed, accuracy, and time-out), it is advised that you move on to the next lesson.

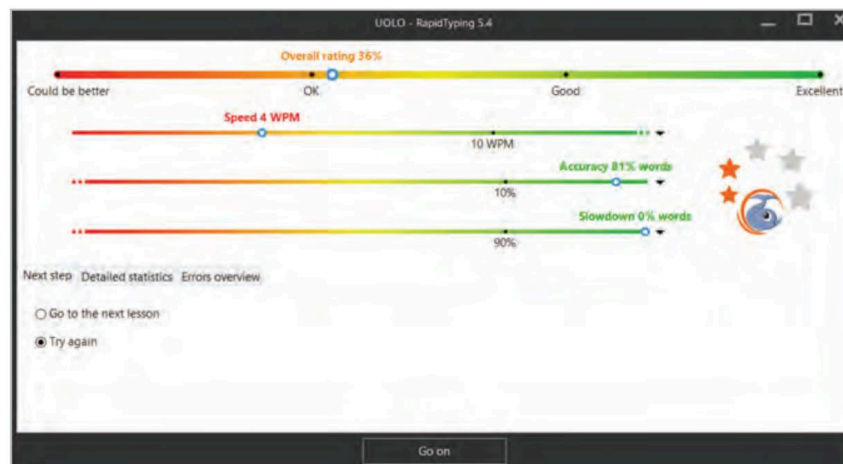


Fig. 3.11. Next Step Tab

Detailed Statistics

You can view the detailed lesson statistics, typing errors and slowdowns for each character/keystroke, and typing speed on the **detailed statistics** tab. It shows the mistakes you made in different colours:

1. The red colour depicts a bad result.

2. The orange colour depicts an incorrect character and the time for typing a character that was exceeded by the user.
3. The yellow colour depicts the time frame that was exceeded by the user.
4. The green dashed line depicts an excellent result.

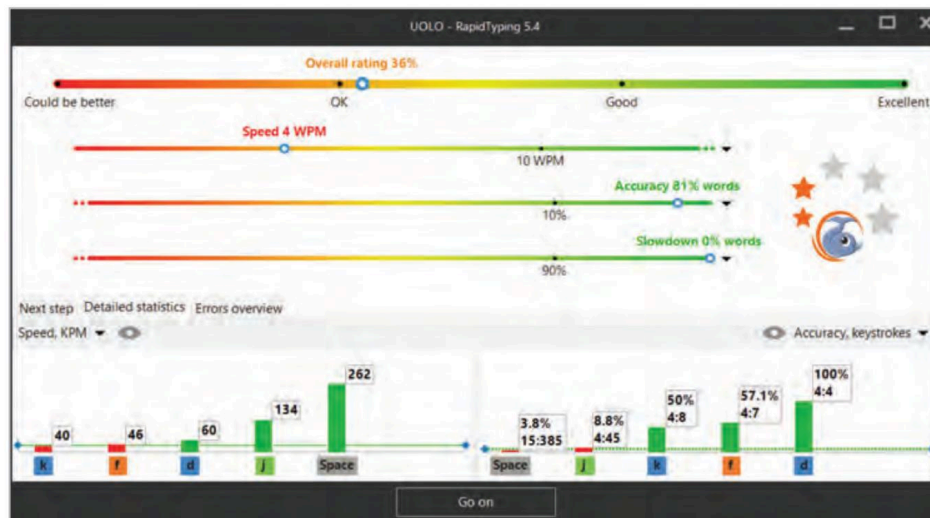


Fig. 3.12: Detailed Statistics Tab

Errors Overview

You can review the errors that occurred during the lesson on the **errors overview** tab.

The colours stand for the following:

1. Green depicts a proper character.
2. A yellow character indicates that the time limit has been exceeded.
3. The characters in red were typed wrong.
4. The time limit for inputting a character has been exceeded, resulting in an incorrect character.

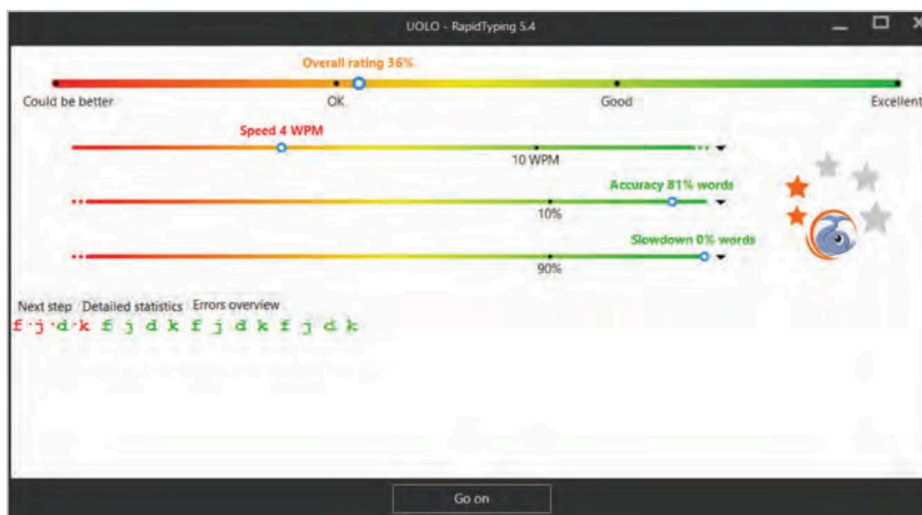


Fig. 3.13: Errors Overview Tab

Remember

These colours can be changed under the background and text colour settings.

Analyse Student Statistics

Select the Student Statistics tab present in the top-left corner of the main window. The screen will open as shown.

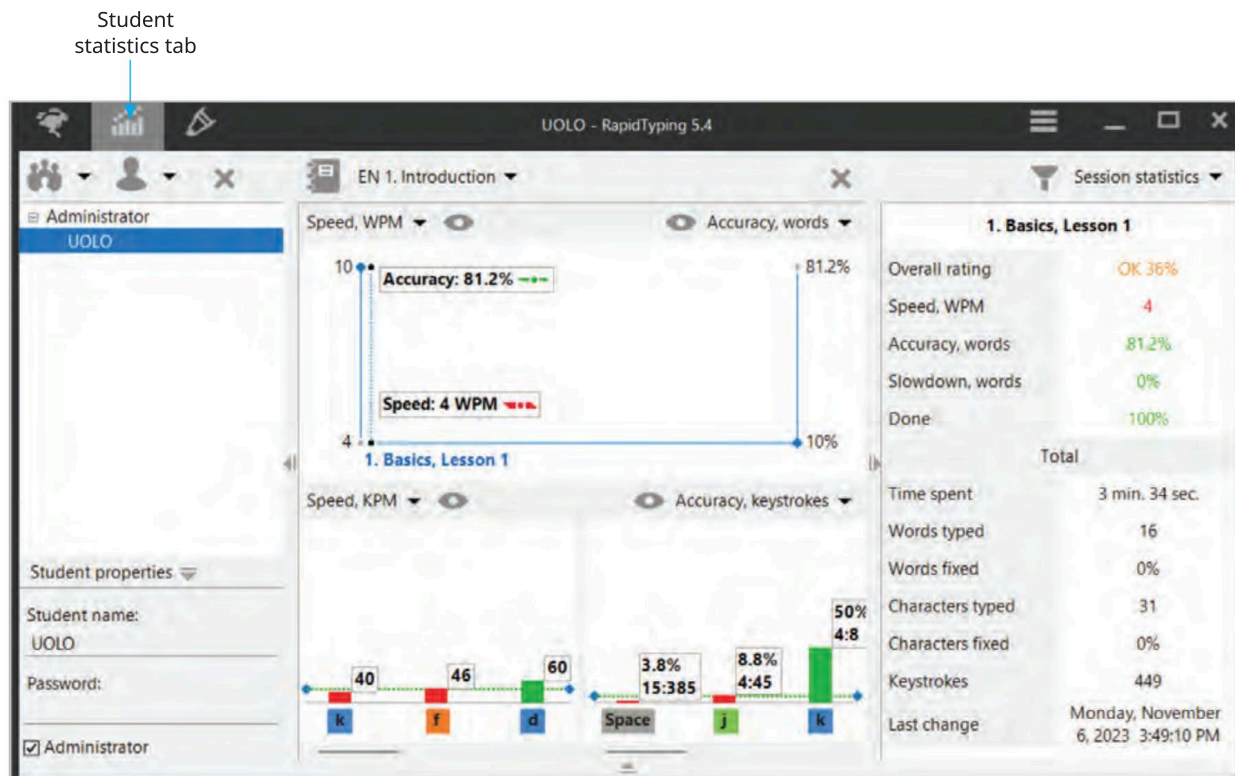


Fig. 3.14: Student Statistics

The student statistics window is composed of the following controls:

1. Student Accounts/Groups: This area has the following parts:

- **Student Toolbar:** It is used for creating new student accounts, editing current student groups, adding new student accounts, deleting existing student accounts, and saving student statistics in the following formats: PDF, HTML, XML, and CSV.
- **Student Accounts:** This section is used for choosing the current student account or group.
- **Property Panel:** You can edit the group or student properties of the currently selected item in the Navigation Tree in the Property panel.

2. Statistics Panel/Session Statistics: This area has the following parts:

- **Statistics Toolbar:** It has the buttons that allow you to pick courses, remove chosen statistics, and customise statistics display choices.
- **Course Diagram/Statistics:** The data for the entire course is displayed in the course statistics. The line moves to red if the outcome is favourable and to green if it is unfavourable.
- **Lesson Diagram/Statistics:** This displays the data for each character or keystroke of the selected lesson in the course diagram.
- **Lesson Statistics Table:** This displays the complete statistics of the selected lesson in the course diagram.

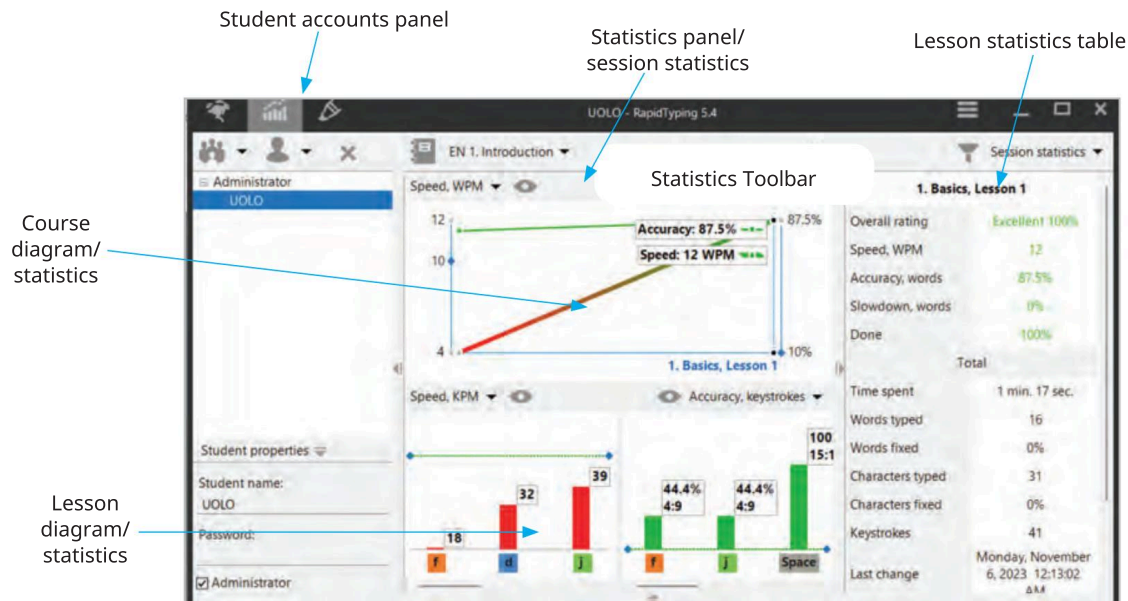


Fig. 3.15: Student Statistics Screen

Working with the Lesson Editor

Lesson editor is used to edit your lessons as per your choice. With a lesson editor, you have complete control over your learning experience. You can create your own courses or alter ones that already exist. You can add, change, delete, or group them together in a course.

To start the lesson editor:

Select the **Lesson Editor** tab present on the top-left corner of the main window. The screen will open as shown.

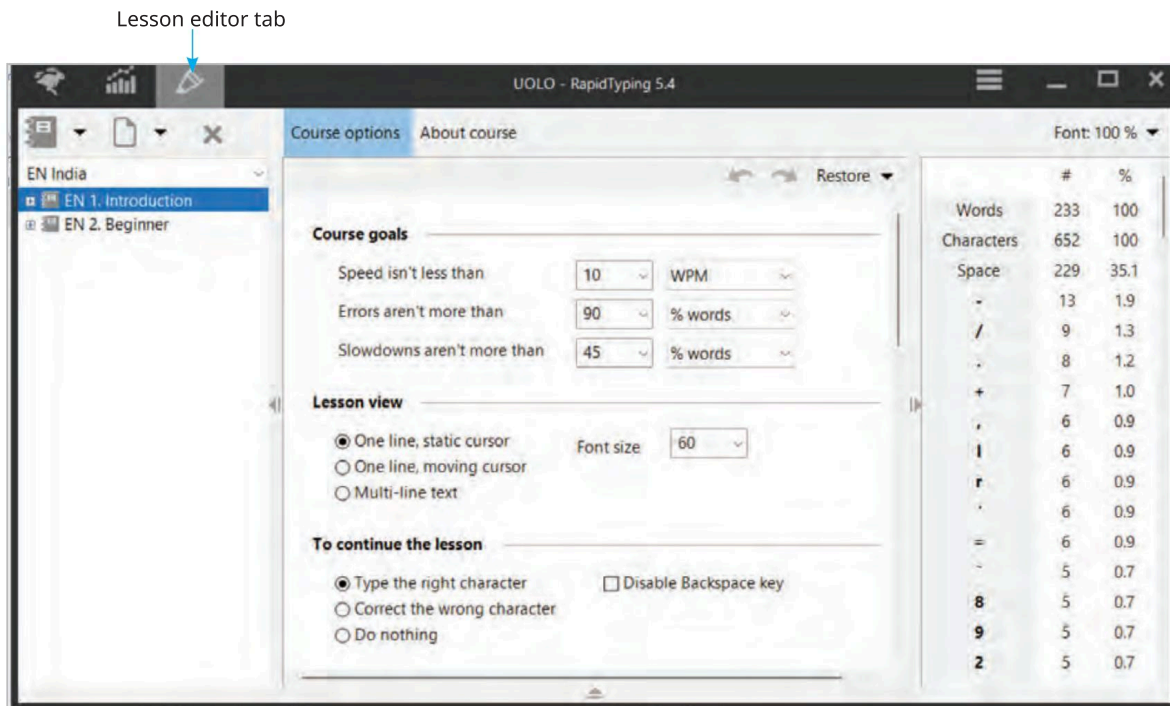


Fig. 3.16: Lesson Editor Window

The controls in the Lesson Editor window are as follows:

Course and Lesson List

Title Bar It shows the lesson editor tab in addition to the title and control buttons.

Toolbar It has commands for adding, removing, inserting, and saving lessons and courses in the database.

Navigation Tree It displays a list of the currently selected score area, lesson, and keyboard configuration.

Properties This section allows you to change the course, section, and properties of the current lesson, including the lesson name and section.

Editing Panel

You can alter the course options or the currently selected lesson in the editing panel, which is located in the navigation tree.

Text Analysis Table

The amount and percentage of words, characters, spaces, and particular characters in the lesson are shown in the text analysis table.

Virtual Keyboard

The characters used in the chosen lesson are highlighted on the virtual keyboard. You can select your keyboard language in the keyboard layout; it depends on the language of the course.

Remember

You can always easily hide/show:

- Click or press Ctrl+K to access the course and lesson list.
- Press Ctrl+L to access the text analysis table.
- Press Ctrl+W to access the properties.
- Press F9 to access the virtual keyboard.

Creating a New Lesson Plan

You can create a new lesson of your choice by following the given steps:

1. Click the **Lesson Menu**.
2. Select the course level where you want to add a new lesson.
3. Choose Create a new lesson from the lesson menu.
4. Give a suitable name and title in the given box.

Creating a New Course

1. Select the **Lesson Editor** option.
2. Click the course menu option, a drop-down menu will appear.
3. Choose the Create new course option. A new course will be added to the navigation tree.
4. Give a suitable name and title in the given box. Press the **Enter** key.

Deleting a Lesson/Course

You can delete the selected lesson or course, using the delete  button or by pressing the **Delete** key on the keyboard.

Activity Time

Activity 1: Typing Practice

(Individual Work)

Anita was practising on RapidTyping software. She typed 240 words in 10 minutes and made mistakes in 6 words. Calculate the following for Anita.

- Gross and Net typing speed
- Word Accuracy Percentage
- Word Error Percentage

Activity 2: Type with Left Hand (Using Letters A S D F)

(Individual Work)

ffffdddddssssaaaaffffdddddssssaaaaffffdddddssssaaa

ff dd ss aa ff dd ss aa ff d ff d ff s ff s ff a ff a

fdsafdsafdsafdsafdsafdsafdsa

asdfasdfasdfasdfasdfasdfasdfasdf

fads dafssafddsaffasdadfssdafafdsfda

Activity 3: Type with Both Hands

(Individual Work)

fad fads lad lads lass alas salad salads dad dads lad lads salads alas

ad add ads adds as ask asks la lad lads lass da dad dada dadasa sad salad

all fall falls alf alfa alfas fad fads salsa ska skald skalds flak flask flasks

Chapter Checkup

A Select the correct option.

- 1 KPM stands for _____.
 - a Keystrokes per minute
 - b Keyboard per minute
 - c Keys per minute
 - d All of these
- 2 In the Rapid typing tutor, which of the following is not true?
 - a Green letters denote correct inputs.
 - b Yellow letters stand for correct inputs in extra time.
 - c Red letters denote wrong inputs within time.
 - d Orange letters indicate wrong inputs within time.
- 3 The text to be typed appears in the _____ in the centre of the window.
 - a status bar
 - b text area
 - c toolbar
 - d background

B Fill in the blanks with the most suitable words.

1. _____ is used to edit the lesson settings.
2. _____ will assist the user in increasing typing speed by learning commonly used syllables and words.
3. _____ will assist you in learning to type with all ten fingers.
4. Gross speed and _____ speed are the metrics used to calculate an operator's typing speed.

C State whether the following is *True* or *False*. Correct the statements that are false.

- 1 In the Rapid Typing tutor, yellow character indicates that the time limit has been surpassed.
- 2 The main components of the Rapid Typing Tutor are the title bar, toolbar, text panel, status bar, and virtual keyboard.
- 3 Typing accuracy is measured as the proportion of wrongly typed characters to total characters.
- 4 Property panel is used for choosing the current student account or group.

D Answer the following questions. (*Solved*)

Q1. What is the colour indication in typing tutor software?

A1. The following is the colour code:

- The red colour depicts a bad result.
- The orange colour depicts an incorrect character and the time for typing a character that was exceeded by the user.
- The yellow colour depicts the time frame that was exceeded by the user.
- The green dashed line depicts an excellent result.

Q2. What does the lesson editor help us to do?

A2. The lesson editor makes all the lessons fully customisable. You can add, insert, change, and delete lessons, group them in courses, and review the corresponding metrics.

Q3. Mahi is unaware of the importance of maintaining a typing rhythm. Explain it to her.

A3. The typing rhythm is highly crucial in touch-typing approaches. Typing rhythm implies that keystrokes should be spaced evenly. The slowdown indicator is used to manage consistent typing speed.

Answer Key

A 1. a 2. d 3. b

B 1. Lesson editor 2. Experienced course 3. Virtual Keyboard 4. net

C 1. True.
2. True.
3. False. Typing accuracy is measured as the proportion of correctly typed characters to total characters.
4. False. The Student account section is used for choosing the current student account or group.

Unit Reflection

Key Terms

Keyboarding Skills: The ability to touch, type, or input data through the keyboard smoothly while typing is called keyboarding skills.

Alphabet Keys: There are 26 alphabet keys on the keyboard. They are marked from A to Z.

Number Keys: There are 10 number keys on a keyboard, numbered from 0 to 9.

Modifier Keys: A modifier key is a key that modifies the action of another key when pressed simultaneously. These are used in combination with other keys to perform any action.

Shift Key: It is used as a keyboard shortcut key in conjunction with another key to switch between uppercase and lowercase letters.

Ctrl Key: To create control characters, combine the control key with other keys. Depending on the program that is running, each control character has a different meaning.

Alt Key: This key, known as an alternate, functions as a second control key. In addition, it is combined with other keys as a keyboard shortcut key.

Navigation Keys: The navigation keys are the keys on a keyboard used to move around a screen.

Caps Lock key: All alphabetic characters are capitalised when the setting is **ON**; otherwise, they are all lowercase.

Scroll Lock key: This key is no longer seen on the majority of contemporary keyboards because it is rarely used. However, if it is available on a keyboard, it aids in stopping page scrolling until it is in use.

Num Lock key: This key is used to turn numeric 1–9 keys off. With this key in on mode, the numeric keys 8, 4, 2, and 6 can be used as navigation keys.

Pointing Device: A pointing device is a sort of input device that allows a user to interact with a computer by moving a pointer across a screen to choose icons and perform desired actions.

Gross Speed: Gross or raw speed is a measurement of how quickly you type without accounting for errors.

Net Speed: Net speed is a calculation of how quickly you can type error-free.

Typing Accuracy: It is measured as the proportion of correctly typed characters to total characters.

Typing Errors: Typing error is the percentage of wrongly typed words out of the total number of typed words.

Things to Remember

- Data entry is the process of transcribing information or entering physical data into an electronic device, such as a computer, through data entry or input devices.
- Touch typing is done using a standard QWERTY keyboard with the hands placed at a starting location called the home row keys.
- There are generally 14 punctuation marks used in the English language.

- The special keys present on the keyboard are used to carry out special actions in the text.
- The top row of the keyboard is home to the function keys, which start with the letter F (F1 to F12).
- The Caps Lock, Num Lock, and Scroll Lock keys are commonly referred to as toggle keys.
- A mouse is a small handheld input device that allows you to move around the screen and enter commands without touching the keys.
- The science of typing ergonomics focuses on utilising good posture and stretching practices when using a computer to prevent frequent occupational injuries.
- Examples of typing software are Rapid Typing Tutor, Typing Instructor, The Typing Cat, Typing Master, etc.
- Rapid Typing Tutor is a free and open-source software (FOSS) program designed to help users improve their typing skills.
- Rapid typing offers a variety of courses based on difficulty level, including Introduction, Beginner Experienced, Advanced, and Testing.
- Typing rhythm implies that keystrokes should be spaced evenly.

Test Your Knowledge

A. Select the correct option.

- _____ is the process of transcribing information or entering physical data into an electronic device.
a. Data entry ☐ b. Touch typing ☐ c. Keyboarding skills ☐ d. Pointing device ☐
- The ability to touch, type, or input data through the _____ smoothly while typing is called a keyboarding skill.
a. Mouse ☐ b. Keyboard ☐ c. Printer ☐ d. Monitor ☐
- The _____ section allows you to start/stop the current lesson and view the current speed, accuracy, and elapsed lesson time in real time.
a. Lesson editor ☐ b. Text area ☐ c. Status bar ☐ d. Lesson control ☐
- _____ is measured as the proportion of correctly typed characters to total characters.
a. Typing speed ☐ b. Typing error ☐ c. Typing accuracy ☐ d. Touch typing ☐
- The _____ dashed line depicts the excellent result.
a. Green ☐ b. Yellow ☐ c. Red ☐ d. Orange ☐

B. Fill in the blanks with the most suitable words.

- _____ has buttons that allow you to pick courses, remove chosen statistics, and customise statistics display choices.
- The _____ colour depicts the incorrect character and the time for typing a character that was exceeded by the user.
- _____ displays the data for each character or keystroke of the selected lesson in the course diagram.
- _____ key is used to give spacing between two independent words to form a sentence.
- A _____ serves as a guide for typing.

C. State whether the following is *True* or *False*. Correct the statements that are false.

- The lesson diagram displays the complete statistics of the selected lesson in the course diagram.
- The home row keys are A, S, D, F, G, J, K, and L.
- The Status bar allows you to customise the primary program features.
- When Num Lock is off, the keys 8, 2, 4, and 6 function as arrow keys for navigation.
- OMR is used to read and input information from a form.

D. Short answer-type questions.

- Define the Status bar.
- What do you mean by typing accuracy?
- What is the use of a barcode reader?

E. Long answer-type questions.

- What are navigation keys?
- Define title bar. Discuss all the buttons placed on it.
- Discuss the allocation of keys to fingers on the third row.

F. Competency-based questions.

1. Pihu wants to know about the available courses while learning Rapid Typing. Explain to her the variety of courses based on difficulty level.
2. Sneha is typing using the keyboard, but not following the proper typing ergonomics. What is typing ergonomics? Explain the importance of sitting posture while working on a computer.